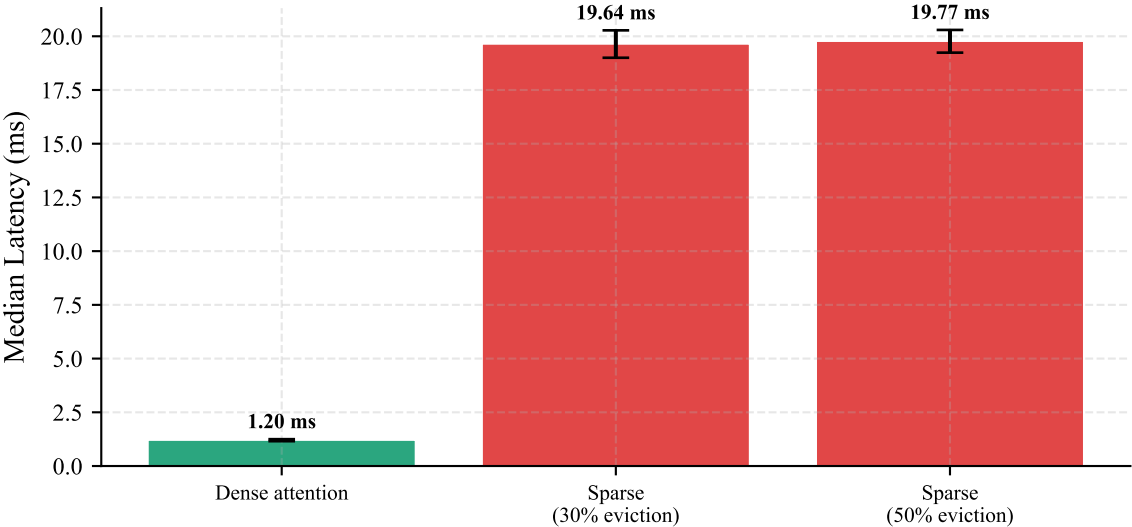


Latency Profiling — Dense vs. OrthoCache Sparse Attention

Kernel Timing — Kaggle TPU v5e-8



Summary Statistics

Metric	Dense	Sparse 30%	Sparse 50%
Median (ms)	1.20	19.64	19.77
P95 (ms)	1.30	21.01	20.96
Std (ms)	0.047	0.638	0.529
Iterations	50	50	50

NOTE: Sparse kernel overhead (16x) is due to prototype Python-level dispatch, not algorithmic cost. Production XLA-fused implementation expected to achieve net speedup at long contexts (>32K tokens).